

PROJECT REPORT

**Title: Predictive Pulse: Harnessing Machine Learning for
Blood Pressure Analysis**

TEAM ID: 699ee17ef292113d819d8e62

Team Members:

- 1.Vanshika Dewan[Team Lead]**
- 2.Chandrasekar TM**
- 3. Archita Singh**
- 4. Palakonda Bhargav Ram**

Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis

PROJECT DESCRIPTION:

- **System Overview:** An advanced machine learning project designed to predict and classify various stages of hypertension based on clinical parameters and lifestyle factors.
- **Problem Domain:** Healthcare technology and predictive medical analytics.
- **Core Technology Used:** Python, Scikit-Learn , Pandas, Numpy, Machine Learning Models(Logistic Regression, Decision Tree, Random Forest, K Nearest Neighbour, XG Boosting), and Flask
- **Key Capabilities:** Automated hypertension classification, risk scoring, and personalized health recommendations.
- **Target Users:** Patients for preventive screening, healthcare providers for decision support, and medical researchers.
- **Real-world Relevance:** Reducing cardiovascular complications through early detection and instant, accessible blood pressure analysis.

APPLICATION SCENARIOS / USE CASES:

- **Hypertensive Monitoring:** Continuous tracking of clinical data to alert patients of shifting blood pressure stages.
- **Preventive Screening:** Used in clinics to identify high-risk individuals before they develop severe symptoms.

- **Emergency Triage:** Assisting medical staff in prioritizing patients based on predicted hypertension severity and risk scores.

TECHNICAL ARCHITECTURE OVERVIEW:

- **Frontend Layer:** HTML and CSS interface for entering patient health data and viewing results.
- **Backend Layer:** A prediction pipeline that handles data normalization, model inference, and recommendation generation.
- **Data / Request Flow:** Data Preprocessing -> ML Model Inference -> Recommendation Module -> Results Display.
- **Deployment Environment:** Localhost for development with potential for secure cloud hosting.

PREREQUISITES:

Software Requirements

- **Development Tools:** VS Code/ Google Collab
- **Runtime Environment:** Python 3.9+

Libraries / Frameworks / Dependencies

- **Packages:** Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn.

PRIOR KNOWLEDGE REQUIRED:

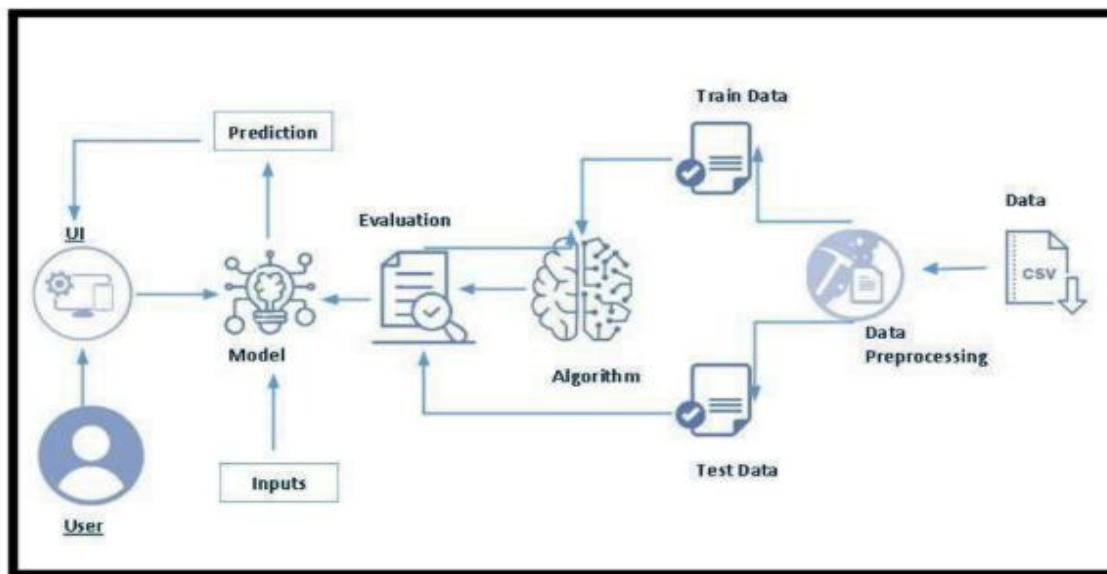
- **Programming:** Strong foundation in Python for data manipulation and backend logic.

- **AI/ML Fundamentals:** Understanding of Supervised Learning, Classification algorithms, and model evaluation.
- **Framework Basics:** Knowledge of Flask for web routing and model deployment.

PROJECT OBJECTIVES:

- **Technical:** Predict and classify stages of hypertension using clinical parameters and lifestyle data.
- **Performance:** Optimize model performance through hyperparameter tuning to achieve high classification accuracy.
- **Learning Outcomes:** Master the end-to-end ML pipeline, from data cleaning to web-based deployment.

SYSTEM WORKFLOW:



1. **User Interaction:** User enters clinical data (age, symptoms, lifestyle habits) into the UI.
2. **Input Handling:** System validates inputs and encodes categorical lifestyle variables.
3. **Processing Logic:** Data is normalized and passed through the prediction pipeline.
4. **Core System Execution:** The supervised learning model performs inference to classify the hypertension stage.
5. **Output Generation:** Results and personalized recommendations are mapped to the user session.
6. **Response Delivery:** User is redirected to a results page to see the diagnosis and risk score.

7. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

- **Problem Definition:** Addressing the need for accurate, accessible hypertension classification using clinical data.
- **Technology Stack Justification:** Python and Scikit-Learn were chosen for their robust libraries for classification and performance evaluation.

-

MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION:

- **10.1 Setup:** Creating a virtual environment and installing required ML libraries via pip install.
- **10.3 Project Structure:** Setting up data, models, templates, and static directories.

MILESTONE 3: CORE SYSTEM DEVELOPMENT:

- **11.1 Feature 1 - Data Preparation:** Cleaning the dataset, handling missing values, performing Exploratory Data Analysis to find correlations and converting categorical values to numerical
- **11.2 Feature 2 - Model Building:** Training multiple algorithm such as Logistic Regression, Decision Tree ,Random Forest ,K Nearest Neighbours , Gradient Boosting, Support Vector Machine and tuning hyperparameters.
- **11.3 Feature 3 - Recommendation Module:** Implementing logic to provide health suggestions based on the predicted stage.

MILESTONE 4: INTEGRATION & OPTIMIZATION:

- **12.1 Integration:** Connecting the trained ML model with the Flask backend and UI.
- **12.2 Optimization:** Selecting the best-performing model based on evaluation results.

MILESTONE 5: TESTING & VALIDATION:

- **13.1 Test Cases:** Validating the system with diverse scenarios (e.g., normal BP vs. emergency triage cases).
- **13.5 Performance Metrics:**
 - **Accuracy:** Overall percentage of correct stage classifications.
 - **Precision/Recall:** Ensuring high reliability in identifying hypertensive cases.
 - **F1 Score:** Balancing precision and recall for medical decision support.

- **Model Training**

Models	Accuracy	Precision	F1 Score
Logistic Regression	0.91	0.90	0.90
Decision Tree	1.00	1.00	1.00
Random Forest	1.00	1.00	1.00
Support Vector Machine	0.95	0.95	0.96
K Nearest Neighbour	0.91	0.91	0.91
Naïve Bayes	0.84	0.88	0.84

Hyperparameter Tuning:

Models	Accuracy	Precision	F1 Score
Logistic Regression	0.91	0.90	0.90
Decision Tree	1.00	1.00	1.00
Random Forest	1.00	1.00	1.00
Support Vector Machine	1.00	1.00	1.00
K Nearest Neighbour	0.99	0.99	0.99
Naïve Bayes	0.84	0.88	0.84

DEPLOYMENT:

- **Hosting:** Initially deployed on Localhost with production-ready ethical disclaimers.

PROJECT STRUCTURE:

- app.py: Main Flask application file
- data: Contains the collected patient dataset(patient.csv)
- models: Saved trained classifiers (Log_model. pkl file).
- templates: HTML files for the web interface.
- requirements.txt: List of dependencies.

ADVANTAGES & LIMITATIONS:

- **Advantages:** High speed, provides actionable health recommendations, and uses accessible clinical data.
- **Limitations:** Decision-support only; not a substitute for a professional medical diagnosis.

CONCLUSION

This implementation successfully harnesses machine learning to provide accurate hypertension analysis, offering a practical impact through early detection and patient-specific risk scores.

APPENDIX

GitHub Repository:

<https://github.com/vanshika-21-dewan/Predictive-Pulse-Harnessing-Machine-Learning-for-Blood-Pressure-Analysis.git>